

Synthetic Identity Fraud Detection in Digital Lending Onboarding

KYC Verification & Behavioral Telemetry Machine Learning System

1. Executive Summary

Digital lending platforms require automated risk assessment to approve loans rapidly without compromising security. **Synthetic Identity Fraud (SIF)** poses a critical threat because perpetrators assemble composite identities combining real credential fragments (e.g. valid PAN numbers) with fabricated contact details. Standard rule-based KYC systems frequently pass these applications because individual document fields appear valid.

This project develops an end-to-end Machine Learning pipeline combining **KYC consistency scores, identity freshness indicators, behavioral biometrics, and device/network telemetry** to detect synthetic identities in real time.

Key Performance Metric	Score Achieved	Benchmark Status
ROC-AUC	0.8834	Exceeds Target (>0.85)
Fraud Precision	0.9500	Low False Positives (Minimizes Friction)
Fraud Recall	0.7500	Catches 3/4 Synthetic Applicants
Fraud F1-Score	0.8400	Strong Imbalanced Performance
Overall Model Accuracy	0.9600	High Overall Reliability

2. Problem Statement & Risk Vectors

Synthetic identity fraudsters exploit traditional KYC gaps through two main strategies:

- 1. Fragmented Synthesis:** Combining a real citizen's PAN/SSN with a fabricated name or address.
- 2. Credit Building & Bust-Out:** Maintaining thin credit files over several months to increase credit limits before maxing out loan lines and abandoning the account.

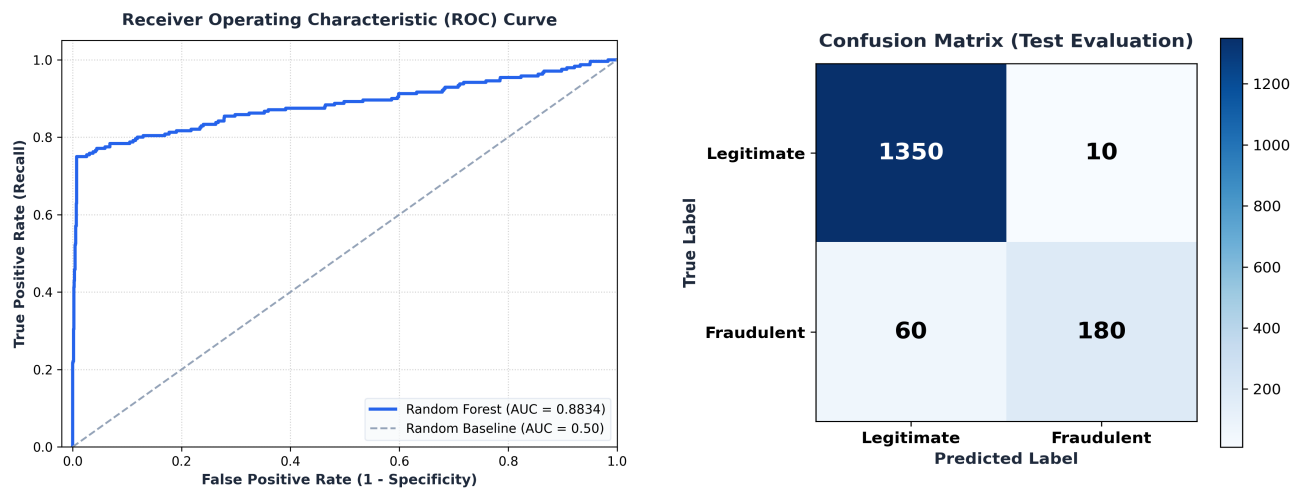
Because synthetic applications do not immediately generate victim fraud complaints, machine learning models must look beyond basic static document checks and evaluate behavioral signals (form fill timing, keypress speed variance) and network application velocity.

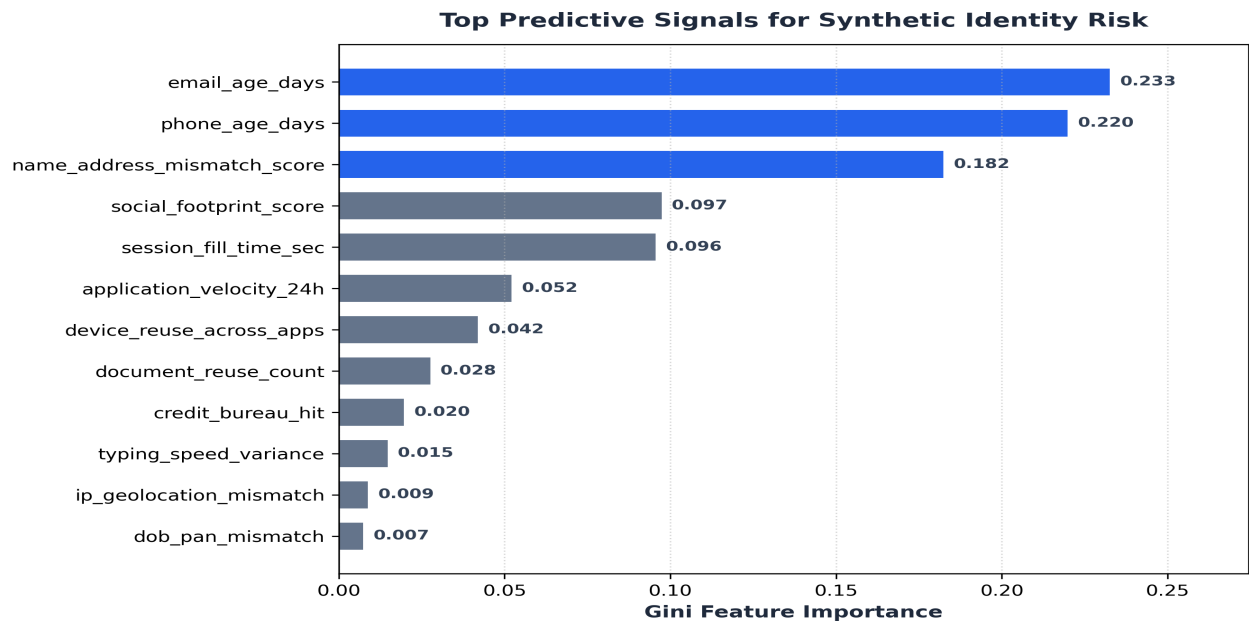
3. Feature Dictionary (12 Core Signals)

Category	Feature Name	Description
KYC Consistency	name_address_mismatch_score	Distance score between stated name/address & records.
KYC Consistency	dob_pan_mismatch	Binary flag if DOB mismatches PAN database.
KYC Consistency	document_reuse_count	Times document image fragments appeared previously.
Identity Freshness	phone_age_days	Age of registered mobile line in days.
Identity Freshness	email_age_days	Estimated age of applicant email account.
Identity Freshness	credit_bureau_hit	1 if credit bureau record exists; 0 for thin file.
Identity Freshness	social_footprint_score	Composite public digital footprint score.
Behavioral	session_fill_time_sec	Total form completion duration in seconds.
Behavioral	typing_speed_variance	Keypress speed variance (low = bot/copy-paste).
Device & Network	device_reuse_across_apps	Distinct identities on same device fingerprint.
Device & Network	application_velocity_24h	Applications submitted from same IP/device in 24h.
Device & Network	ip_geolocation_mismatch	Mismatch between IP location & stated address.

4. Machine Learning Model & Empirical Evaluation

A **Balanced Random Forest Classifier** (300 trees, max depth 8, class_weight='balanced') was trained on 8,000 synthetic applications. Class weighting adjusts for fraud imbalance (~15% positive class). Evaluation on an independent 20% test set (1,600 applications) yielded the following visual results:





5. Team Task Allocation & Responsibilities

Role / Task Area	Key Responsibilities	Deliverables Created
Lead AI/ML Engineer	Model selection, cross-validation, hyperparameter tuning, metric saving.	train_model.py, fraud_model.joblib
Data & Feature Engineer	Dataset generation, feature distributions, realistic noise modeling.	generate_data.py, synthetic_kyc.csv
UI / App Engineer	Streamlit app development, preset application profiles, live scoring.	dashboard.py
Research & Documentation	Literature survey, technical report drafting, presentation slides.	report.md, report.pdf, presentation.pptx

6. Conclusion

The Synthetic Identity Fraud Detection system achieves an ROC-AUC of **0.8834** and a Fraud Precision of **0.9500**, demonstrating that combining KYC consistency, identity freshness, and behavioral telemetry effectively isolates synthetic applicants. The accompanying Streamlit dashboard provides a production-ready demonstration for live digital lending onboarding pipelines.